Lecture - 07

$$(V, \oplus, \odot, \#)$$

Set

Linear Combination, $S = \{v_1, v_2, \ldots v_n\}$, $v_i \in V$

Q1: Any fixed $v \in V$

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = v$$

Linear Independence

Q2: $\alpha_1 v_1 + \cdots + \alpha_n v_n = \vec{0}$

$$(\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0)$$

Span

Q3: Does our set S generate the whole vector space?

Motivation:

$$S = \left\{ \overset{w}{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}, \overset{w}{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} \right\} \quad V = \mathbb{R}^2$$

$v_{random} \in \mathbb{R}^2$

$$\begin{pmatrix} a \\ b \end{pmatrix}$$

$$a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$\begin{pmatrix} a \\ b \end{pmatrix}$ $\qquad \alpha \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = \underbrace{v_{random}}$$

$$S = \{\underbrace{v_1, \ldots, v_n}\}$$

$$\underbrace{\mathrm{Span}(S)} = \left\{ \underbrace{\alpha_1 v_1 + \cdots + \alpha_n v_n}, \ \alpha_i \in \mathbb{F} \right\}$$

free

Does $\mathrm{Span}(S) = \underline{\underline{V}}$? If yes, then we say

$\boxed{S \ \underline{\text{span}} \ V.}$

<u>Example</u> $\quad S = \left\{ \underbrace{\begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}} \right\}$

$\mathrm{Span}(S) = \mathbb{R}^2$?

$v_{random} = \begin{pmatrix} a \\ b \end{pmatrix}$

$$\alpha \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\begin{pmatrix} 2\alpha \\ 0 \end{pmatrix} + \begin{pmatrix} \beta \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \gamma \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\left.\begin{aligned} 2\alpha + \beta + 0 &= a \\ 0 + 0 + \gamma &= b \end{aligned}\right\}$$

$$\begin{array}{ccc} \alpha & \beta & \gamma \end{array}$$

$$\begin{pmatrix} 2 & 1 & 0 & a \\ 0 & 0 & 1 & b \end{pmatrix}$$

$\#$ free $= \#$ variables $- \#$ pivots

$= 3 - 2 = 1$

$$\beta = t$$

$$\gamma = b$$

$$2\alpha + \beta = a \Rightarrow 2\alpha = a - t$$

$$\boxed{\alpha = \frac{a-t}{2}}$$

$$(\alpha, \beta, \gamma) = \left(\frac{a-t}{2}, t, b\right)$$

$\boxed{\text{Span}(S) = \mathbb{R}^3}$. Hence, S span $\mathbb{R}^3$.

→ Unique/Many: span ✓

→ No solution: Not span ✗

Example

$u = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ $\quad v_{\text{rank}} \in \mathbb{R}^3$

$u = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ $\qquad V_{random} \in \mathbb{R}^3$

$w = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$$\boxed{\alpha u + \beta v + \gamma w = V_{random}}$$

$$\alpha \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\left.\begin{aligned} \alpha + \beta + \gamma &= a \\ \alpha - \beta + 0\cdot\gamma &= b \\ 2\alpha + 2\beta + \gamma &= c \end{aligned}\right\}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 1 & -1 & 0 & b \\ 2 & 2 & 1 & c \end{array}\right) \sim \begin{array}{c} \begin{array}{ccc} \alpha & \beta & \gamma \end{array} \\ \left(\begin{array}{ccc|c} 1 & 1 & 1 & a \\ 0 & -2 & -1 & b-a \\ 0 & 0 & -1 & c-2a \end{array}\right) \end{array}$$

$$\boxed{\gamma = -c + 2a \qquad \beta = ? \qquad \alpha = ?}$$

Unique

$$\begin{array}{c} \gamma \\ \left(\begin{array}{ccc|c} 1 & 1 & 1 & * \\ 0 & -2 & 1 & * \\ 0 & 0 & 0 & * \end{array}\right) \end{array}$$

$0 \neq *$

$0\alpha + 0\beta + 0\gamma = *$

No Solution

Basis

$S = \{v_1, v_2, \cdots \cancel{v_n}\}$

Q4: $S' = \{v_1 \cdots\cdots v_{n'}\}$

$\text{Span}(S') = V$

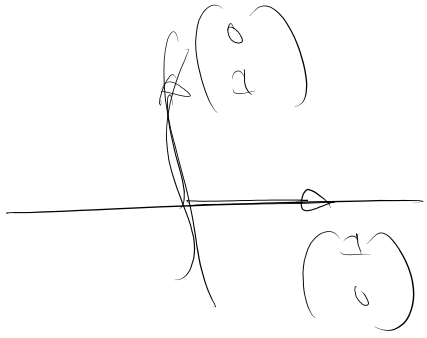
$\underbrace{\text{Span}(S)} = \underbrace{V}$

$\boxed{\text{Minimum Spanning set}}$

Motivation: $\boxed{v_1 + v_2 = v_3}$?

$S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \cancel{\begin{pmatrix} 1 \\ 1 \end{pmatrix}} \right\}$ $\mathbb{R}^2$

$\text{Span}(S) = \mathbb{R}^2$



1. $\text{Span}(S) = V$
2. Independent

Example

$$S = \{ \underbrace{(2,-3,1)}_{v_1}, \underbrace{(4,1,1)}_{v_2} \}$$

$\underline{\underline{\mathbb{R}^3}}$, $M_{2\times 2}(\mathbb{R})$

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

<u>Step 1</u> (Independence)

$$\alpha v_1 + \beta v_2 = \underline{0}$$

$$\alpha(2,-3,1) + \beta(4,1,1) = (0,0,0)$$

$$\Rightarrow (2\alpha + 4\beta, -3\alpha + \beta, \alpha + \beta) = (0,0,0)$$

$$\left.\begin{aligned} 2\alpha + 4\beta &= 0 \\ -3\alpha + \beta &= 0 \\ \alpha + \beta &= 0 \end{aligned}\right\}$$

$$\left(\begin{array}{cc|c} 2 & 4 & 0 \\ -3 & 1 & 0 \\ 1 & 1 & 0 \end{array}\right) \overset{R_1' = \frac{R_1}{2}}{\sim} \left(\begin{array}{cc|c} 1 & 2 & 0 \\ -3 & 1 & 0 \\ 1 & 1 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 7 & 0 \\ 0 & -1 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{array}\right)$$

$$\begin{array}{cc} \alpha & \beta \end{array}$$

$$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & 0 \end{array}\right)$$

$$\boxed{\beta = 0}$$

$$\left(\begin{array}{cc|c} 1 & 2 & \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right) \qquad \boxed{\begin{array}{l} \beta = 0 \\ \alpha = 0 \end{array}}$$

Step 2

$$\alpha v_1 + \beta v_2 = v_{random}$$

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & 7 & b \\ 0 & -1 & c \end{array}\right) \sim\sim\sim \begin{array}{c} \begin{array}{cc} \alpha & \beta \end{array} \\ \left(\begin{array}{cc|c} 1 & 2 & * \\ 0 & 1 & * \\ 0 & 0 & * \end{array}\right) \end{array}$$

$\text{span}(S) \neq \mathbb{R}^3$ No solution

Hence, S is not a Basis of $\mathbb{R}^3$.

Example

$$M_{2\times 2}(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \; a, b, c, d \in \mathbb{R} \right\}$$

$$e_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad e_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad e_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$e_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Basis

Basis

$$S = \{e_{11}, e_{12}, e_{21}, e_{22}\}$$

$$e_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$V = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \; a+b+c+d = 0 \right\}$$

$$\begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$$

Discard

Example (step 2) Indepen.

$$\alpha v + \beta u + \gamma w + \delta x = 0$$

$$\alpha(t^3 - 2t^2 + 4t + 1) + \beta(2t^3 - 2t^2 + 9t - 1) + \gamma(t^3 + 6t - 5) = 0$$

$$t^3(\alpha + 2\beta + \gamma) + t^2(-2\alpha - 2\beta + 0\gamma) + t(4\alpha + 9\beta + 6\gamma) +$$

$$\alpha - \beta - 5\gamma = 0t^3 + 0t^2 + 0t + 0$$

$$\left.\begin{aligned} \alpha + 2\beta + \gamma &= 0 \\ -2\alpha - 2\beta + 0\gamma &= 0 \end{aligned}\right\}$$

$$\left.\begin{aligned} -2\alpha - 2\beta + 8\gamma &= 0 \\ 9\alpha + 9\beta + 6\gamma &= 0 \\ \alpha - \beta - 5\gamma &= 0 \end{aligned}\right\}$$